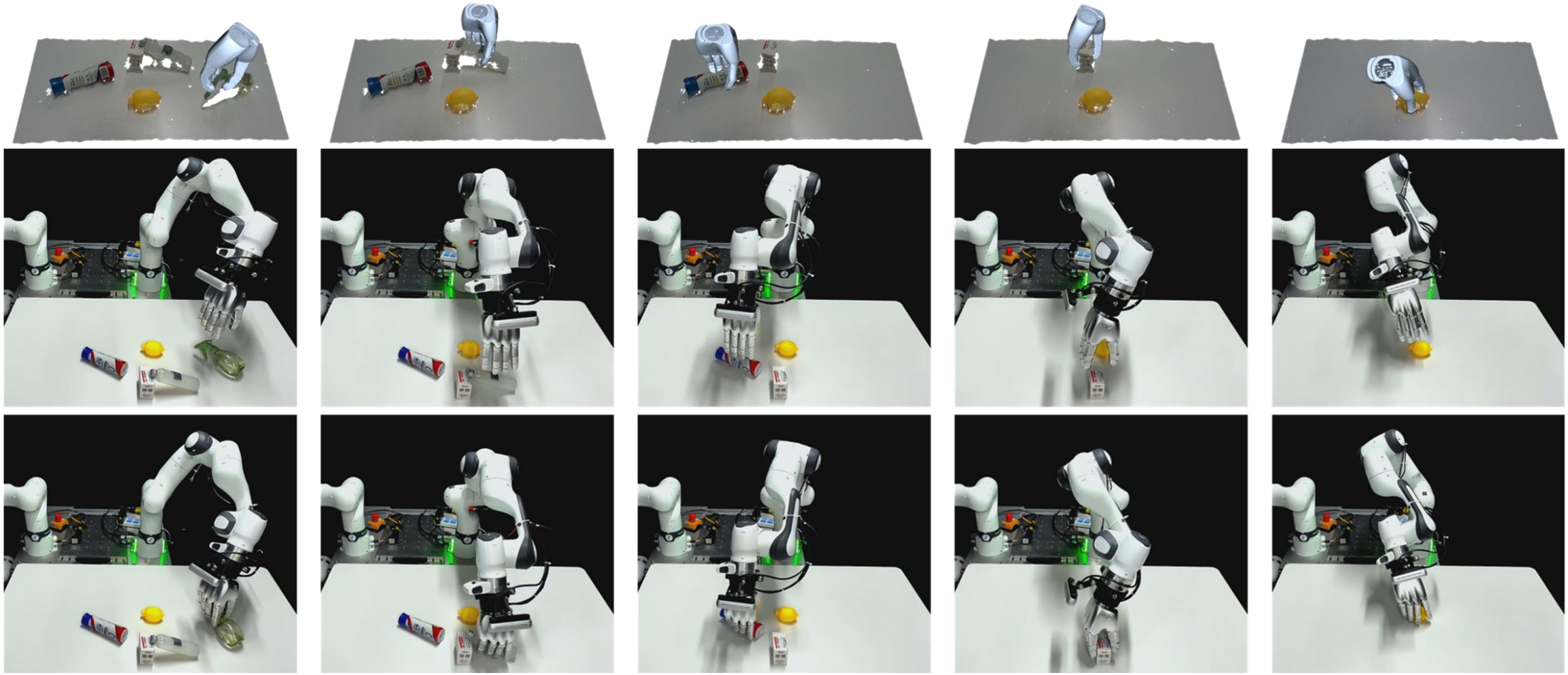
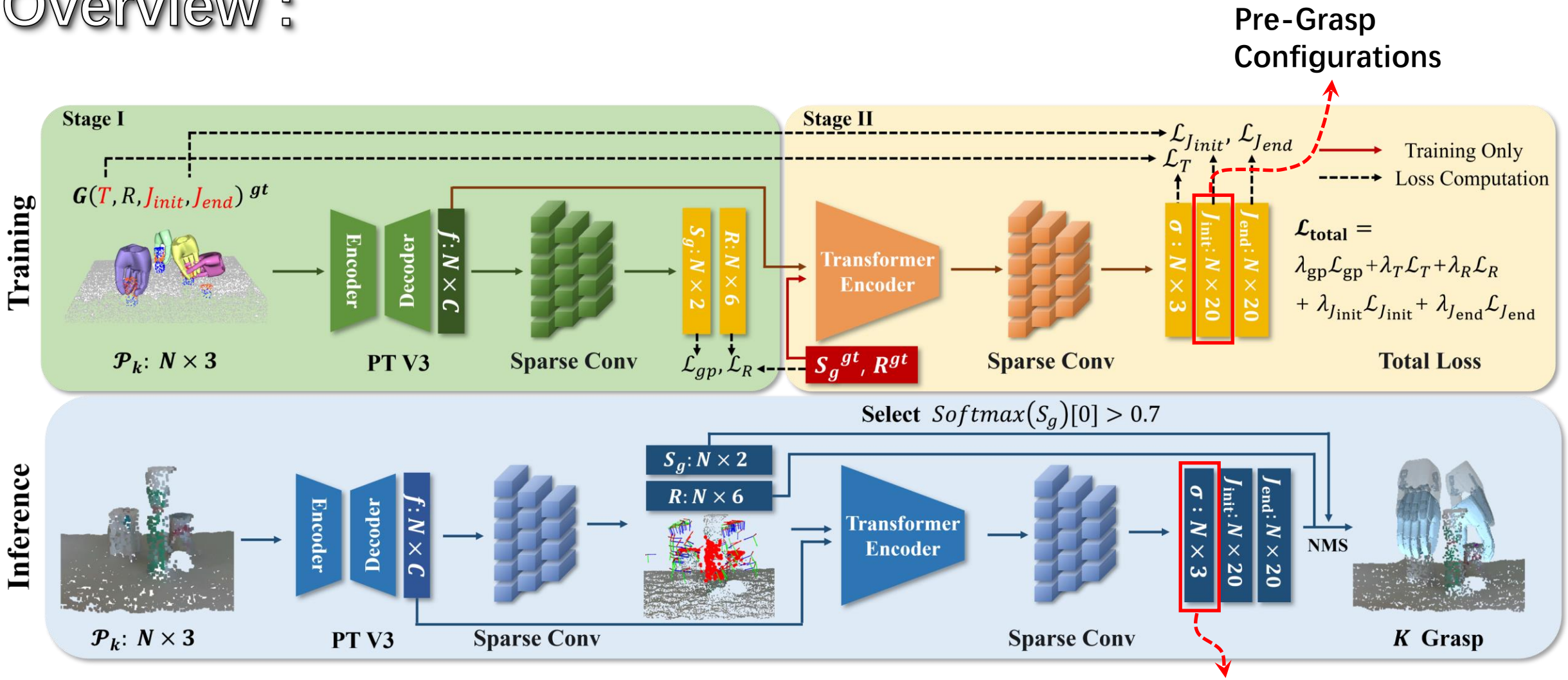
DGS-Net : Learning Dexterous Grasps from Single-View Point Clouds via a Multi-Object Scene Dataset



Overview :



Finally grasp hand's TCP :

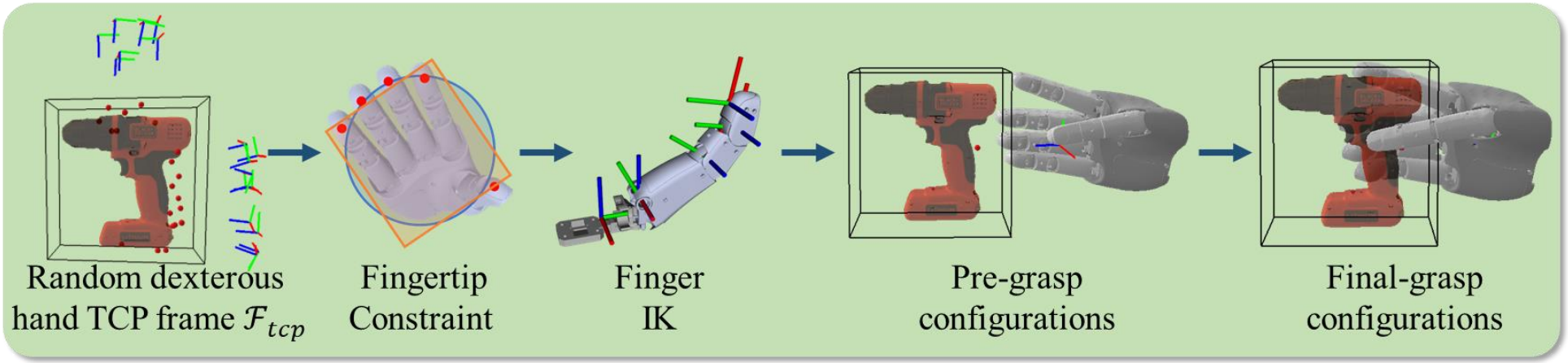
$$T = P_{G_s} + \sigma$$

Module I (green) :
predicts grasp reference point and 6D rotations;

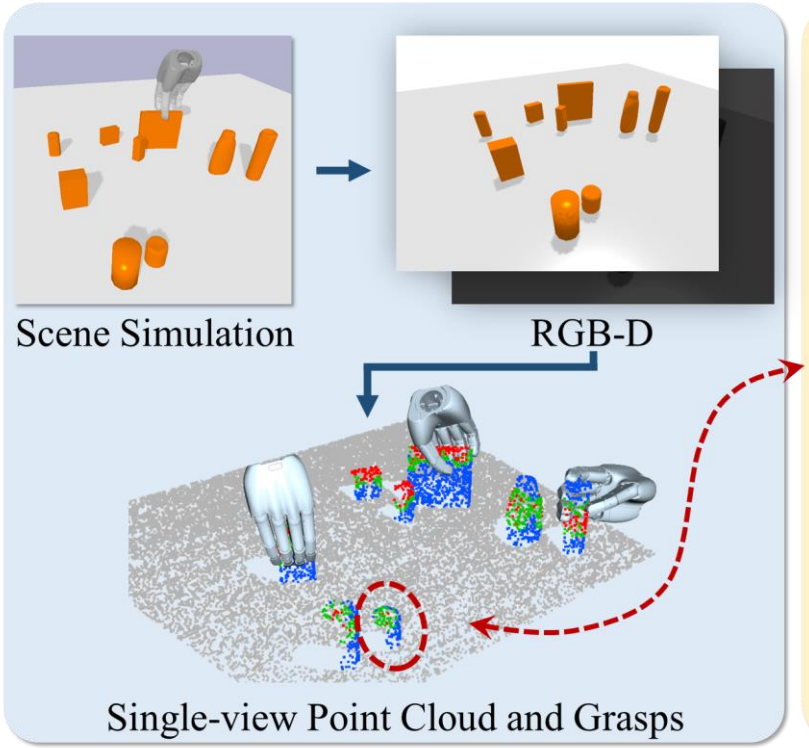
Module II (yellow) :
generates grasp offsets and joint configurations.

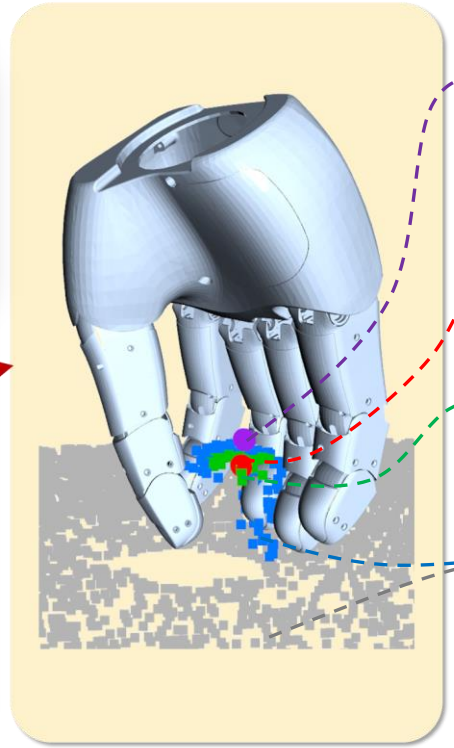
DataSet :

Single-Object Grasp Dataset



Scene-Level Grasp Dataset





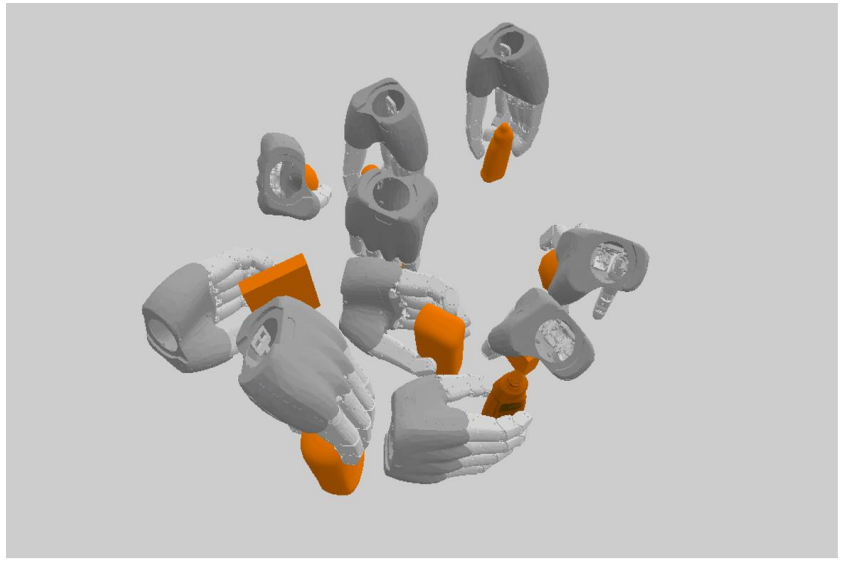
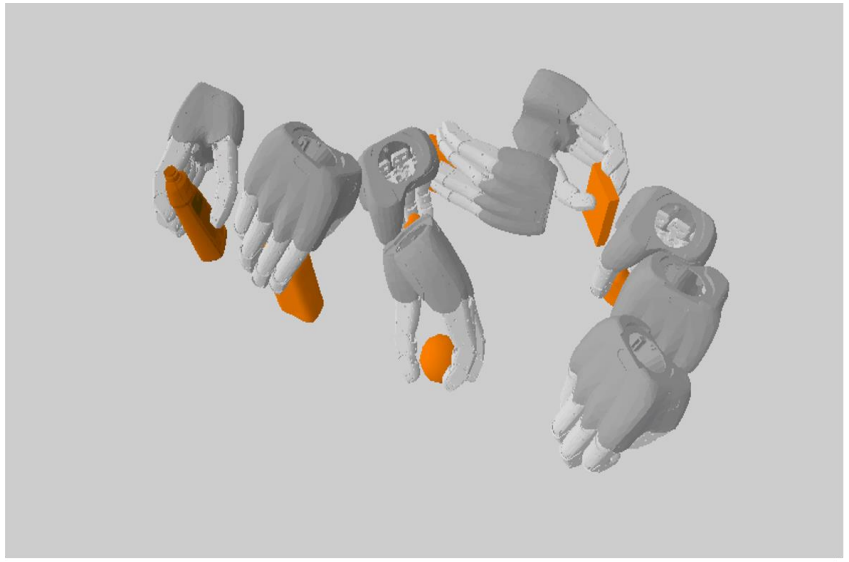
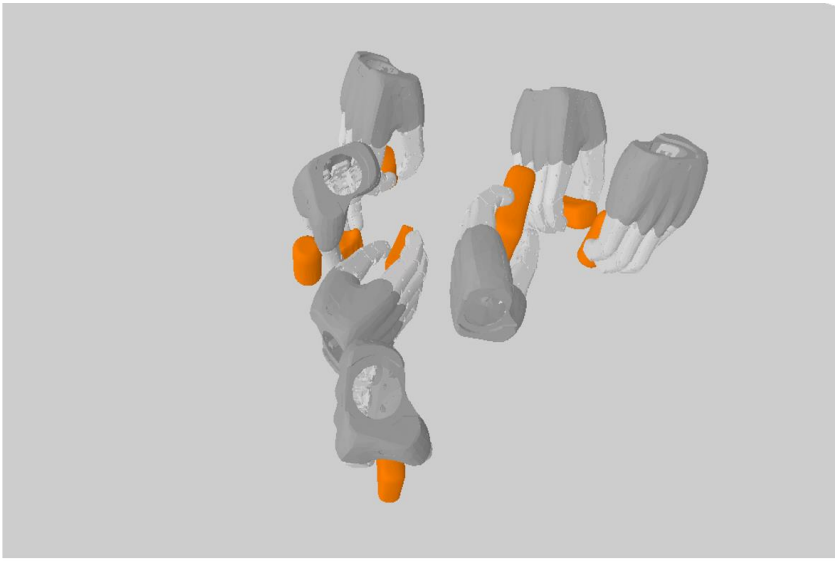
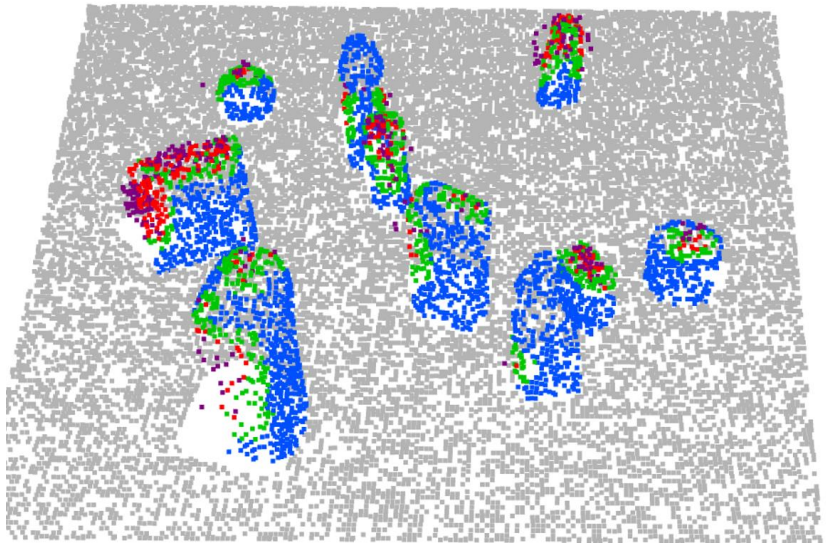
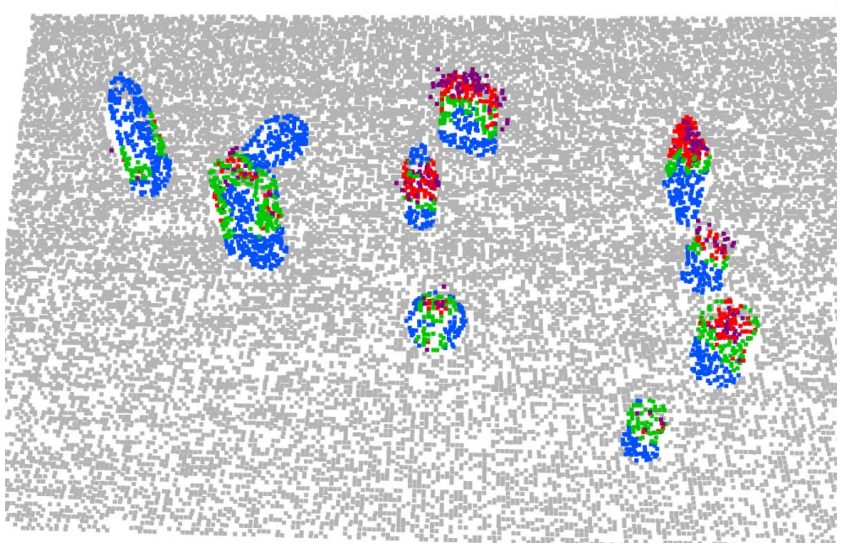
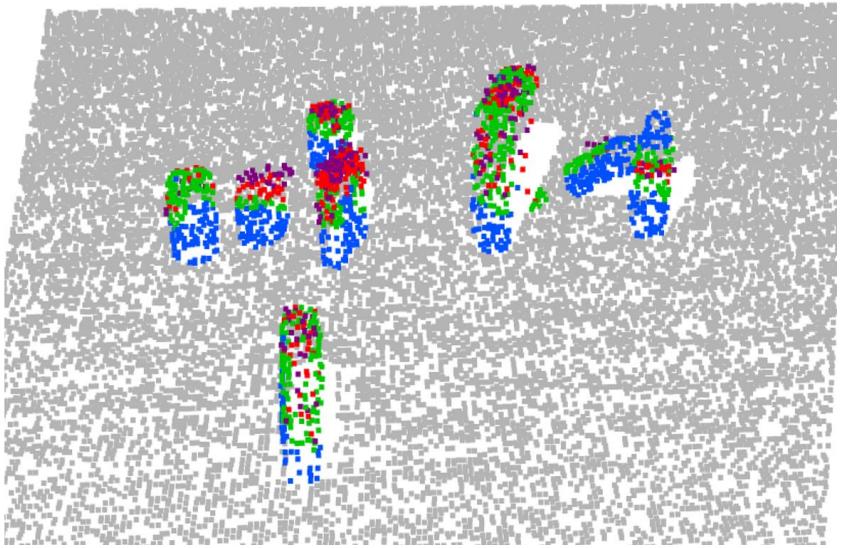
Dexterous Hand TCP:
The TCP of the dexterous hand in the single-object grasp dataset.

Grasp Reference Point:
The point in the point cloud that is closest to the dexterous hand TCP.

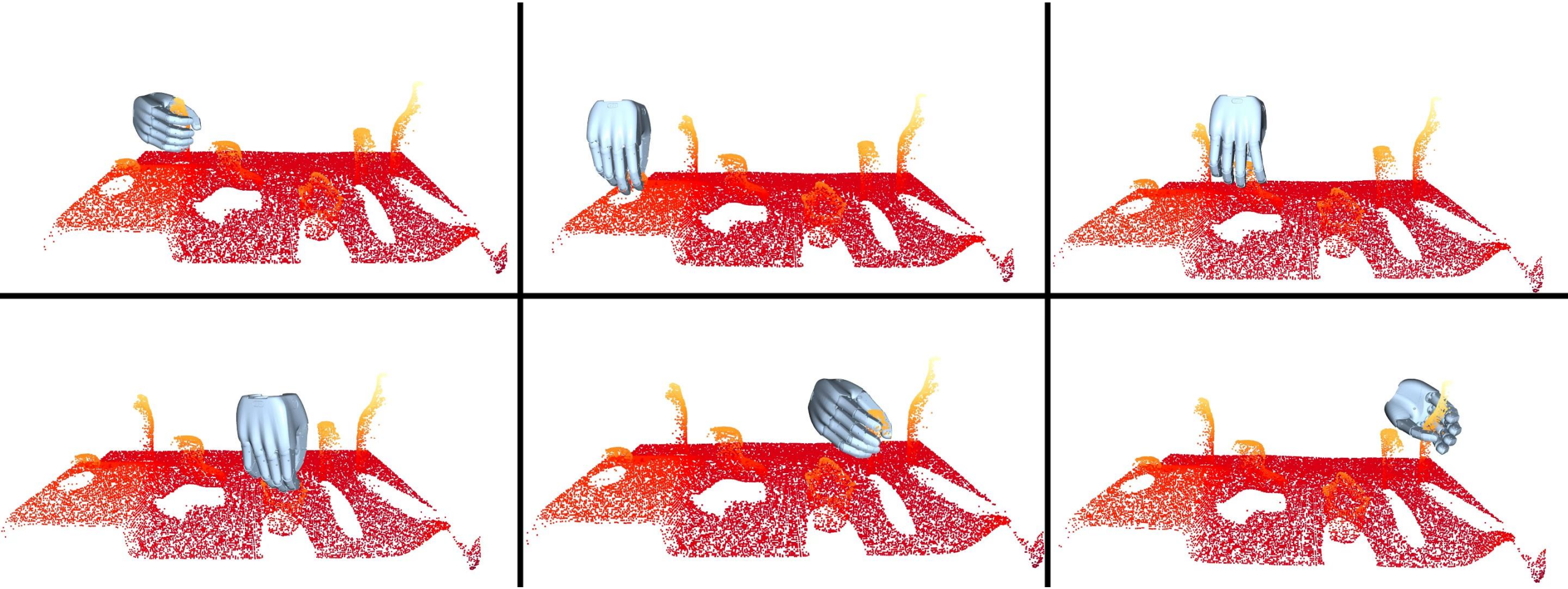
Grasp Candidate Points:
Top-20 nearest points to the hand TCP (excluding the reference point).

Non-Grasp Points:
The remaining object points and the desktop points.

DataSet :

Experiment :

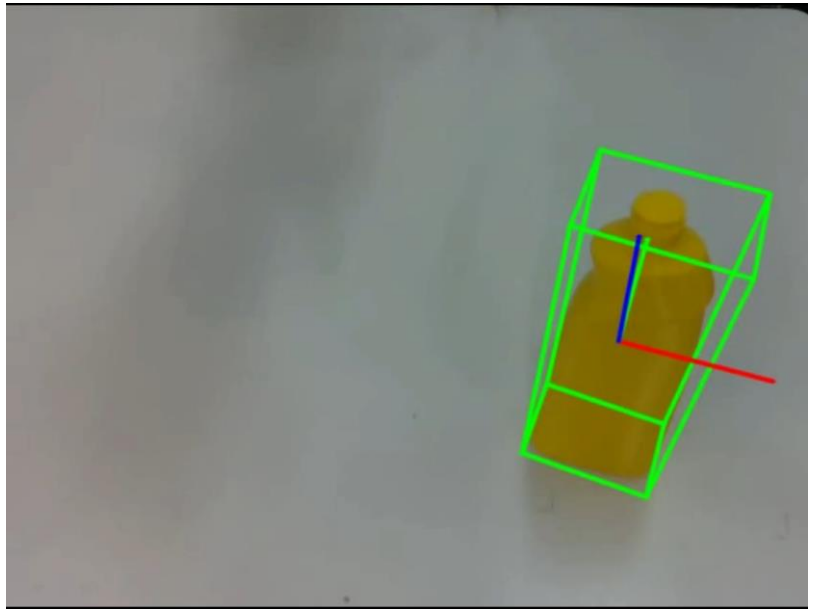


Grasp Success Rate : 79.39%

Scene Clearance Rate : 87.94%

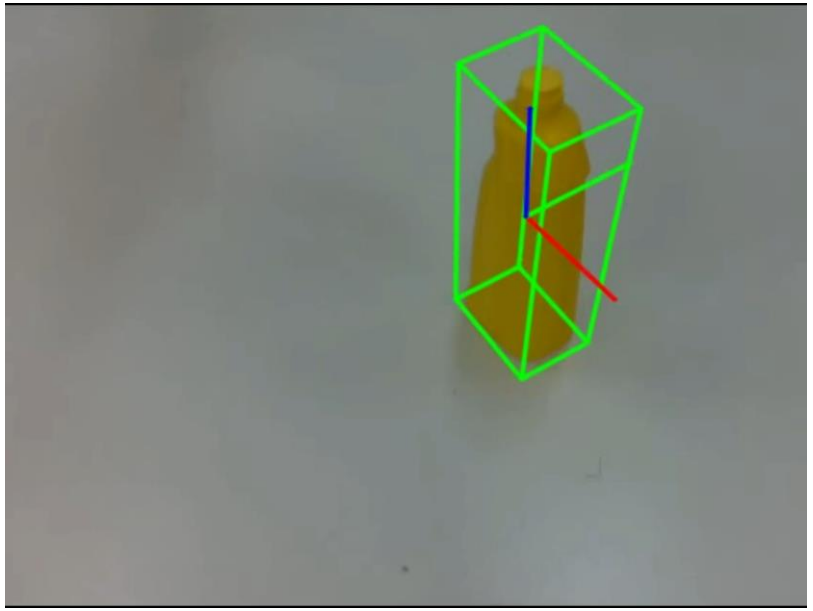
Experiment :

HGC-Net†



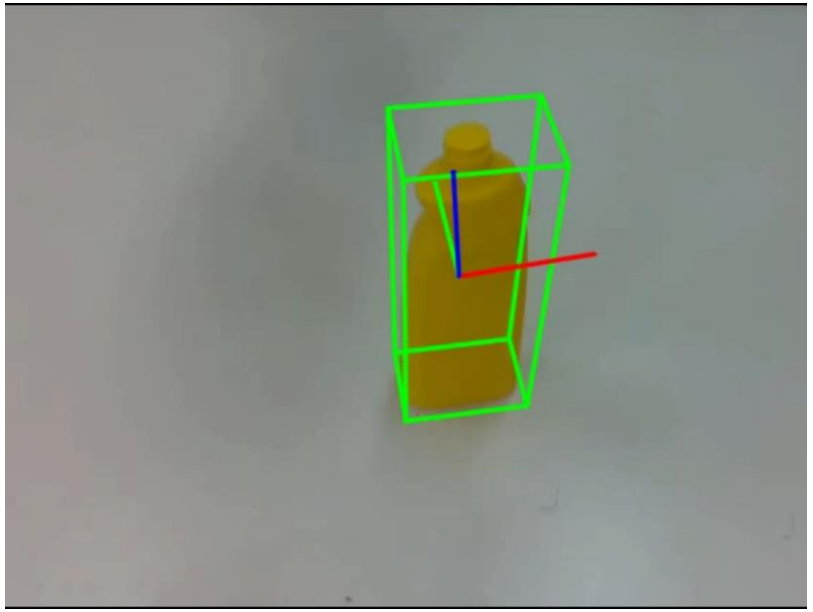
Rot. Error : 29.1070°
Trans. Error : 44.2863mm

DexGraspNet 2.0†

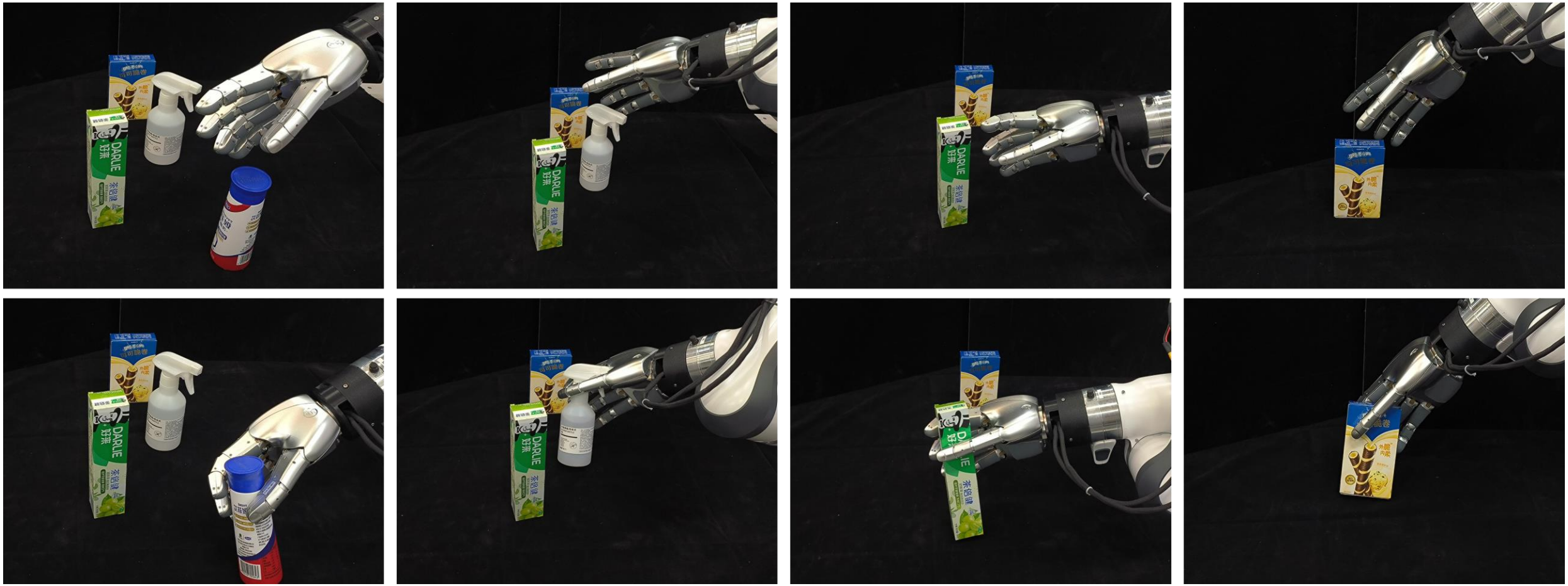


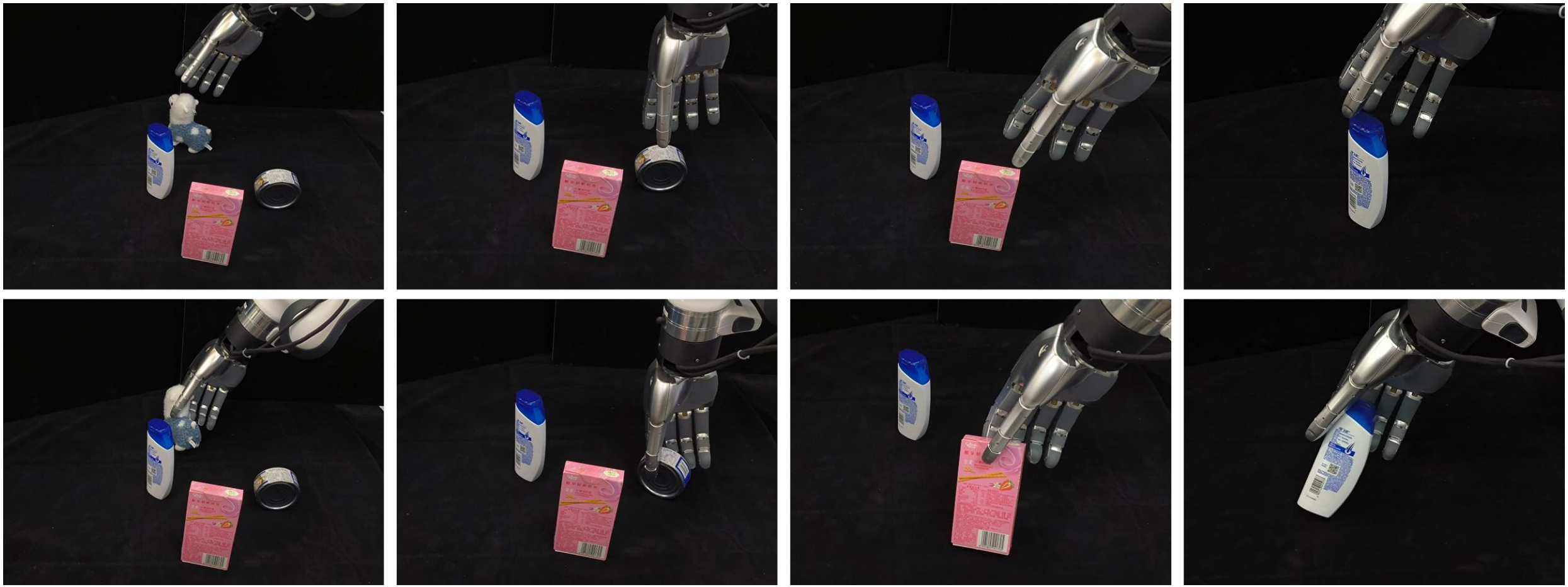
Rot. Error : 10.2332°
Trans. Error : 28.0283mm

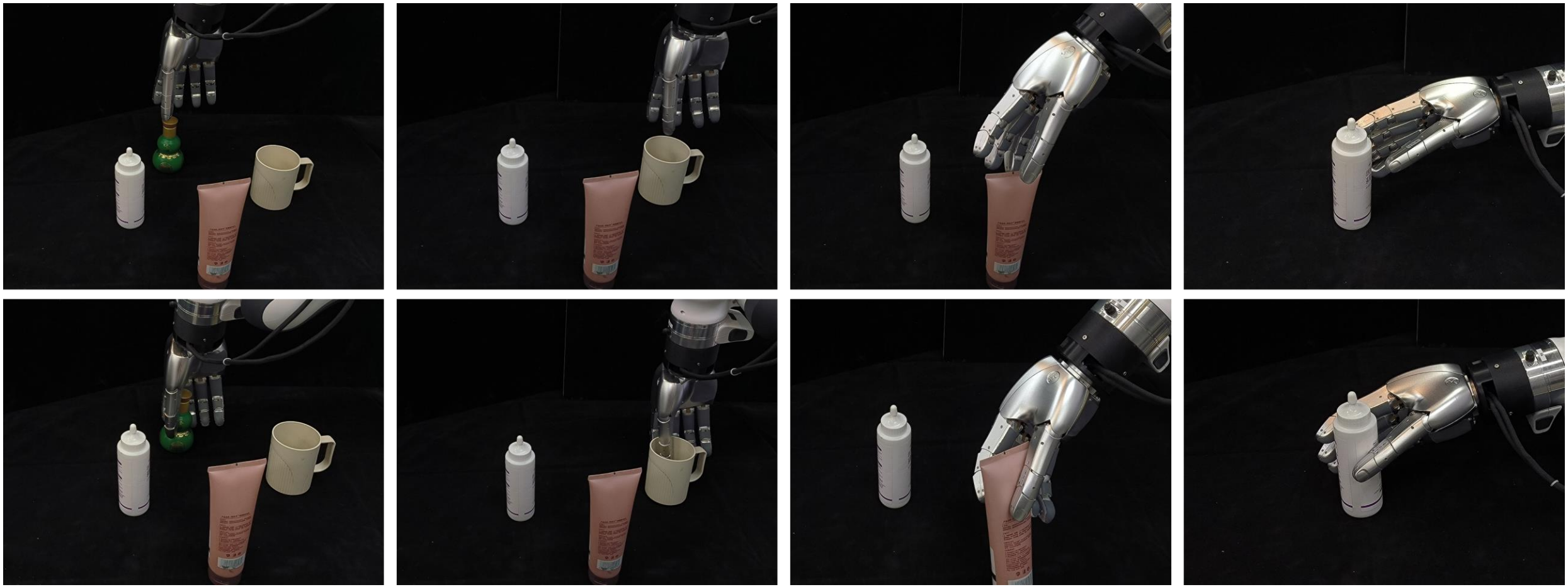
Ours

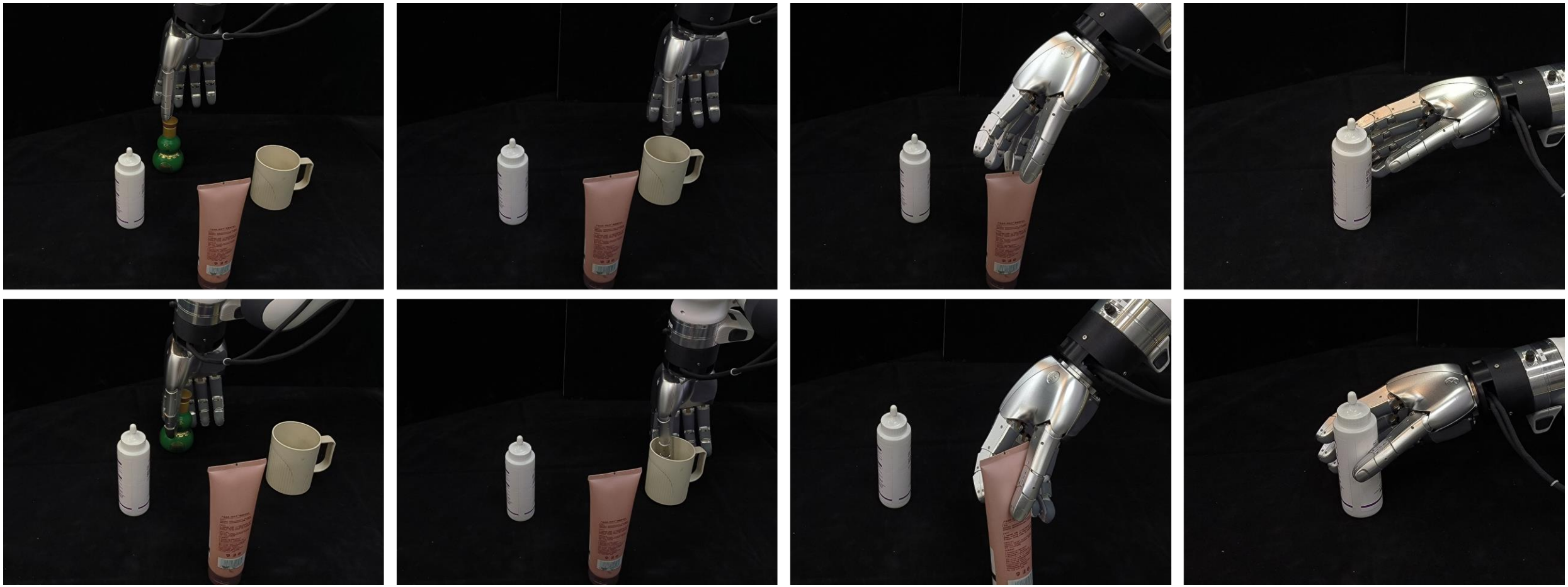


Rot. Error : 3.7436°
Trans. Error : 13.9598*mm*


DARLIE
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茶倍健








HGC-Net
GraspTTA
Dexonomy
DexGraspNet 2.0
Ours